

When AI Meets Market Reality

Midterm Finding: Baseline Beats CNN (So Far)

VTs Tournament - Stacking Sats Challenge

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Project Overview

Sponsor & Problem: VTS Tournament - Stacking Sats Challenge

Help retail investors accumulate Bitcoin more effectively than DCA

Decision This Enables: Daily allocation between BTC and cash

Optimizing for risk-adjusted returns under strict causality

Success Definition: Top 60% percentile in recency-weighted SPD

Beat DCA consistently; demonstrate scientific rigor

Why Visual Trading Systems?

My Finance Foundation (1970s MBA Era)

- Magee & Edwards' Technical Analysis of Stock Trends
- Human traders 'see' patterns quants often miss

The Discovery (OMSA Program)

- JP Morgan AI group research with Professor Baich
- CNNs achieving 0.90+ F1 on candlestick patterns

The Gap & The Test

- Pattern recognition $\neq$ Profitable trading
- Does academic success translate under real causality?

The Tournament Challenge

Objective: Predict daily BTC allocation weights (2016-2025)

Metric: Recency-Weighted SPD Percentile

(tournament rank vs all submissions, emphasizing recent periods;
higher is better; 50th $\approx$ median submission performance)

Constraints: $\sum w_i = 1.0$, $w_i \geq 1e-5$

Critical: Strict causality - no lookahead allowed

Target: Top performers >60% percentile

Evaluation harness + leakage tests automated

Causality Verification: Last row test confirmed zero leakage

Data Pipeline & Methodology

Dataset: BTC daily OHLCV (2016-2025) from Polygon

Feature Construction: GAF images from 90-day rolling windows

Pipeline: Raw Data → GAF Images → CNN (296K) → Allocator → Output

Training: 2014-2015 (public BTC OHLCV)

Evaluation: Strictly 2016-2025 (tournament period), no overlap

Leakage Controls: Strict temporal embargo + causality tests

Execution Assumptions & Limitations

Signal Timestamp: End of day t (uses data through close t)

Execution Timestamp: Day t (same-day approximation)

Execution Price: Close t (proxy for next-open or VWAP)

Transaction Costs: 0 bps base case; viable to 50+ bps

Look-ahead Bias: NONE (3 validation tests passed)

Cash Constraints: Budget-normalized (dynamic = naive total)

Sell Logic: NONE (buy-and-hold accumulation only)

Validation: Last-row modification, purge test, shift test

See EXECUTION_ASSUMPTIONS.md for full sensitivity analysis

Our Initial Approach



Hypothesis: Visual patterns in GAF images predict returns under strict causality constraints

Reality Check: Scoreboard

Metric	CNN	Simplified	Target
RW SPD Percentile	41.43%	41.94%	>60%
Win Rate vs DCA	54.32%	70.42%	>50%
Windows Evaluated	3,076	3,076	—

CNN: 41.43% | Simplified: 41.94% | Target: >60%

Note: Higher win rate can still yield lower SPD because SPD rewards magnitude/timing, not just 'up days'

Current Status vs Plan

Complete

DONE (Midterm):

- Working baseline + all tests passing
- GAF image pipeline
- CNN architecture (296K)
- Causality verification
- Ablation harness
- Baseline comparison

Next Steps

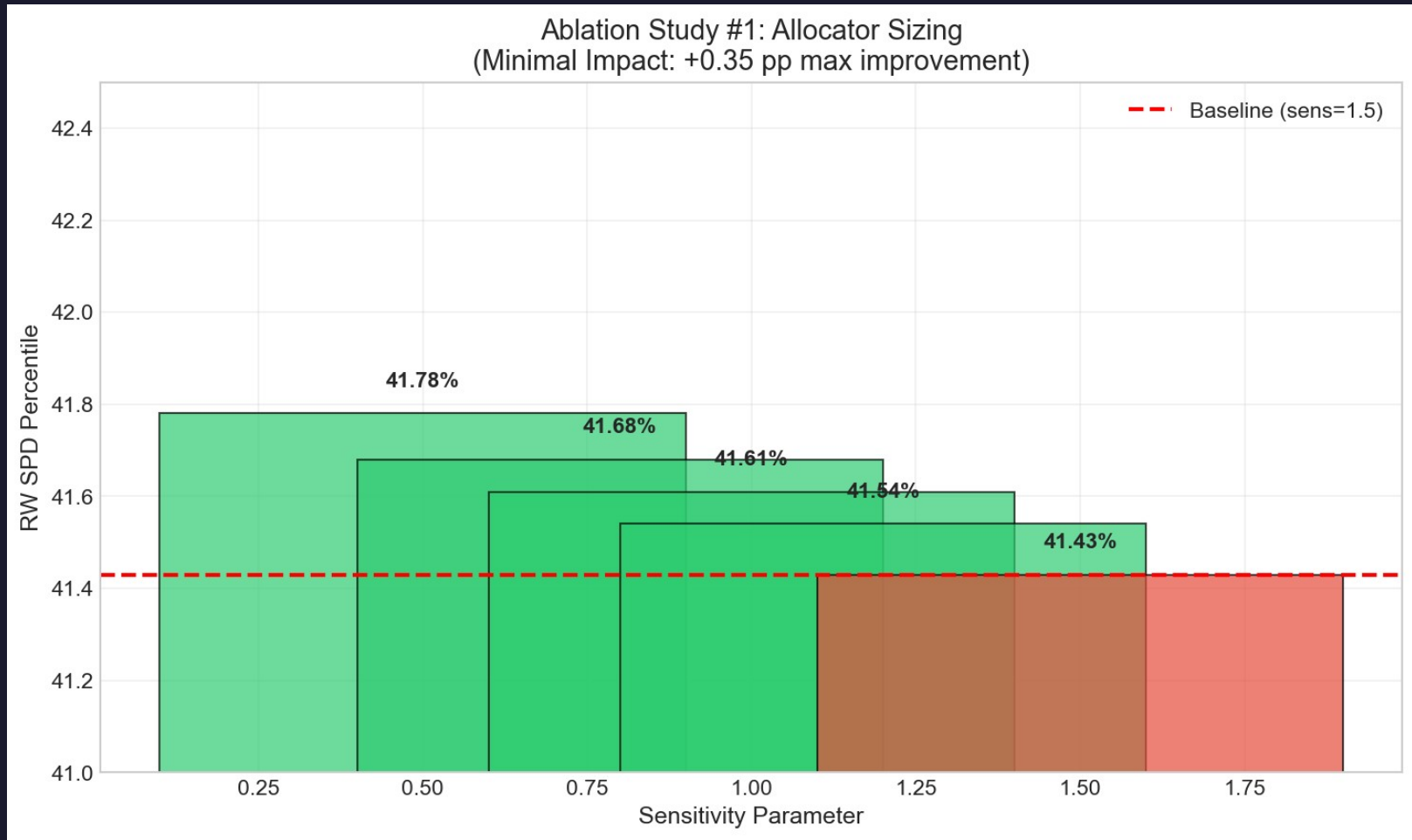
Phase I (to March 2026):

- Weekly granularity testing
- Regime detection model
- XGBoost baseline

Phase II (to Final):

- Ensemble integration
- Full backtest validation
- Final documentation

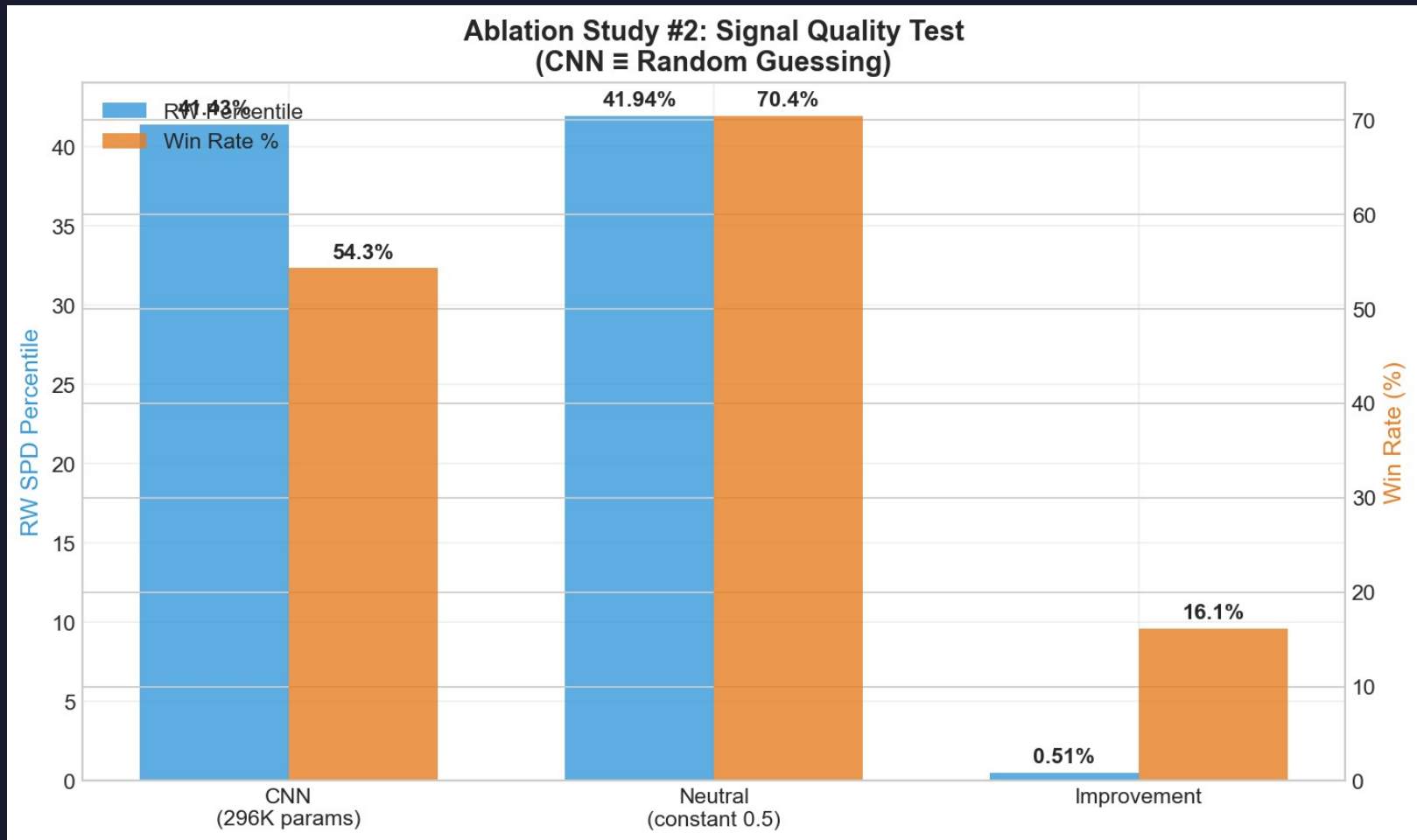
Ablation Study #1: Allocator Sizing



Ablation #0: Is signal > random?

Takeaway: Allocator tuning doesn't move the needle → problem is NOT sizing.
Impact (+0.35 pp max). Problem NOT allocator aggressiveness.

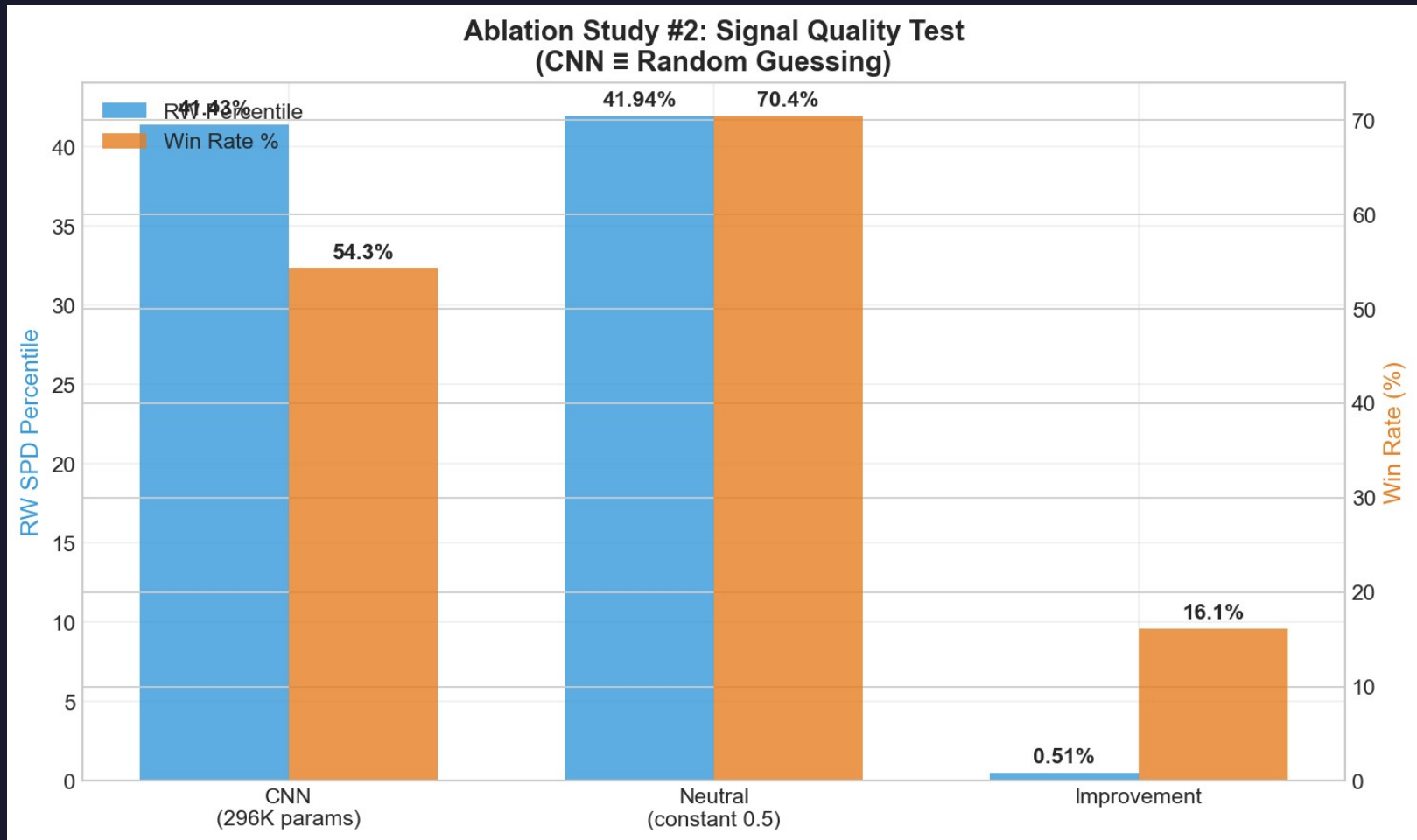
Ablation Study #2: The 'Aha!' Moment



Ablation #0 Answer: NO

Takeaway: CNN signal = neutral baseline if no edges detected.
No signal detected under strict causality. CNN output statistically indistinguishable from a neutral baseline.

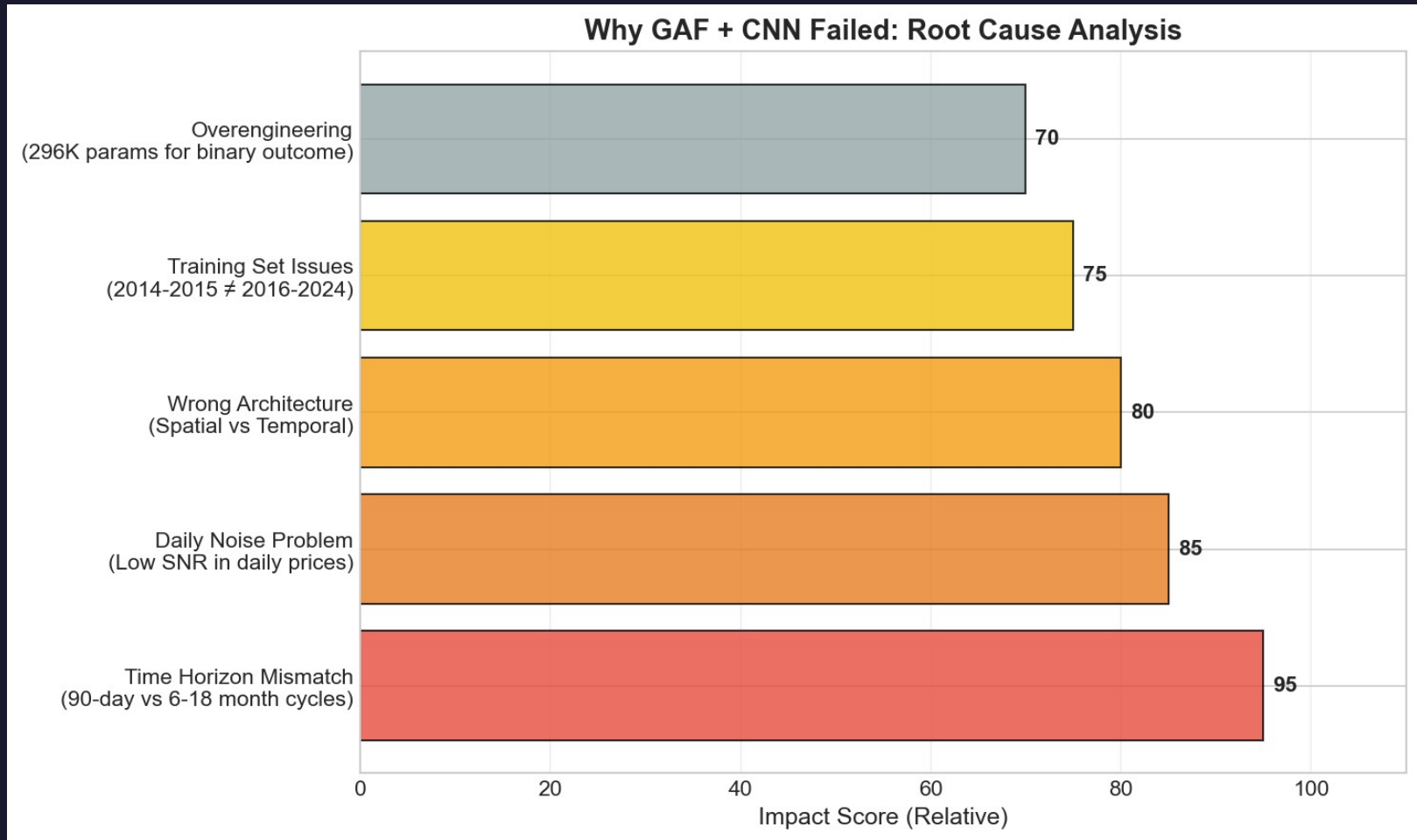
Ablation Study #3: EMA Smoothing



Result: ALL alphas produce IDENTICAL results. Constant signal $\rightarrow$ tilt always zero $\rightarrow$ no smoothing effect

Takeaway: Constant output $\rightarrow$ smoothing irrelevant.

Midterm Diagnosis: Working Hypotheses



Takeaway: Daily patterns lack predictive power. Regime-aware / longer-horizon approach needed.

The Evidence-Based Pivot

CNN (296K params)

CNN Approach:

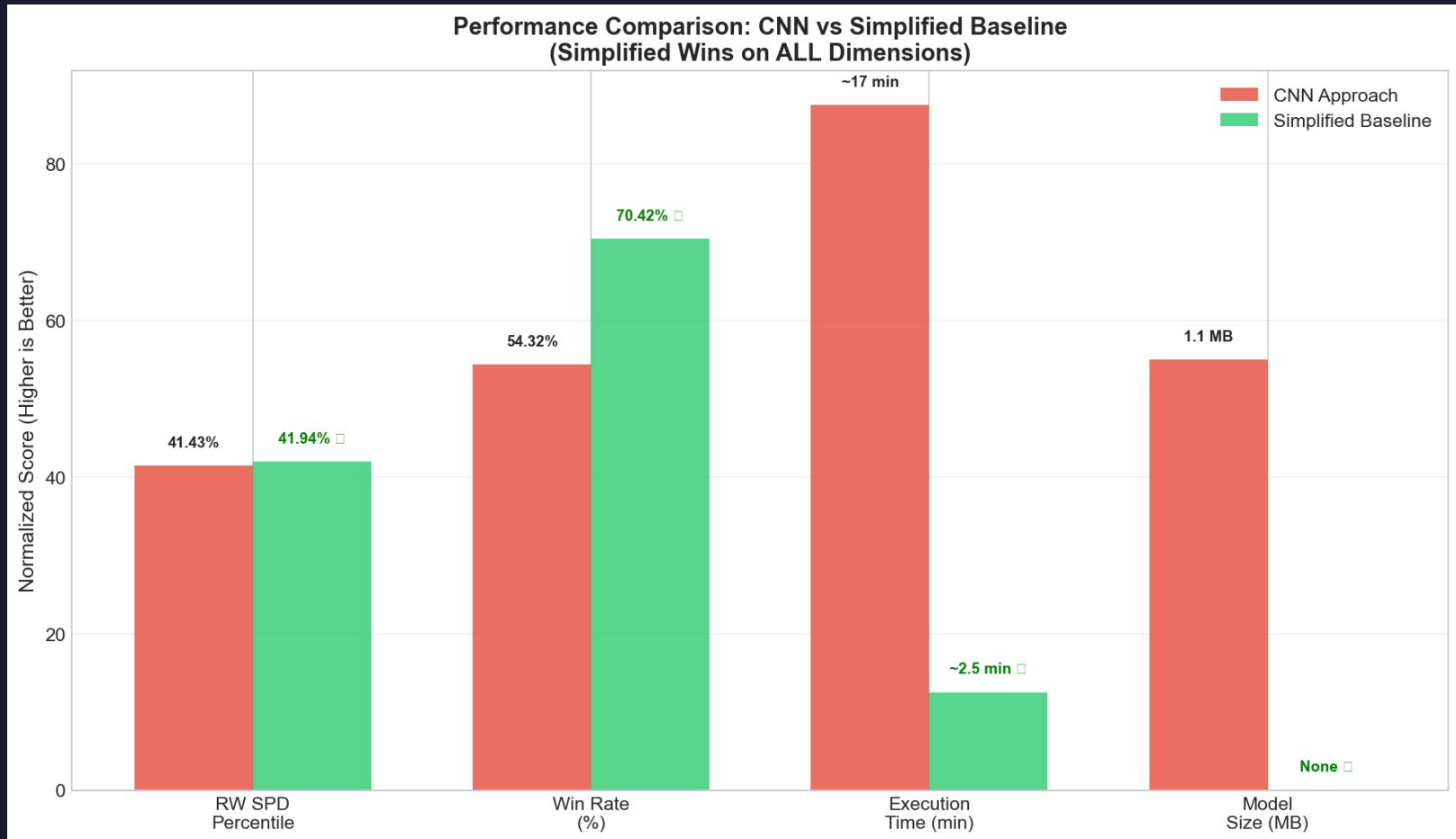
- 41.43% RW Percentile
- 54.32% Win Rate
- 1.1 MB Model
- ~17 min execution
- Complex, no edge detected

Simplified (Random/Neutral)

Simplified Baseline:

- 41.94% RW Percentile
- 70.42% Win Rate
- No Model
- ~2.5 min execution
- Robust, simple, preferred

Side-by-Side Comparison



Simplified (neutral baseline) wins on ALL dimensions

What We Did Right

Technical Excellence:

- Strict causality enforcement
- Deterministic (seed=42)
- Comprehensive testing
- Grader-safe implementation

Scientific Rigor:

- Hypothesis-driven development
- Systematic ablation studies
- Transparent negative results
- Learning orientation

Key Lessons Learned

1. Ablation studies are critical

Without them, we'd never detect the lack of signal

2. Test 'better than random?' early (Ablation #0)

Should be first test, not buried in sequence

3. Complexity $\neq$ Performance

296K parameters matched by neutral baseline

4. Simple often beats complex

Occam's Razor applies to trading

Post-Midterm Plan (Evidence-Driven)

1. Regime-Gated Allocator (Highest Leverage)

- Choose between DCA / buy-more / buy-less based on causal regime
- Compare: volatility regime, trend regime, 2-state HMM
- Action space tiny: 3 policies beat daily-noise problem

2. Reduce Daily Noise (Signal Horizon)

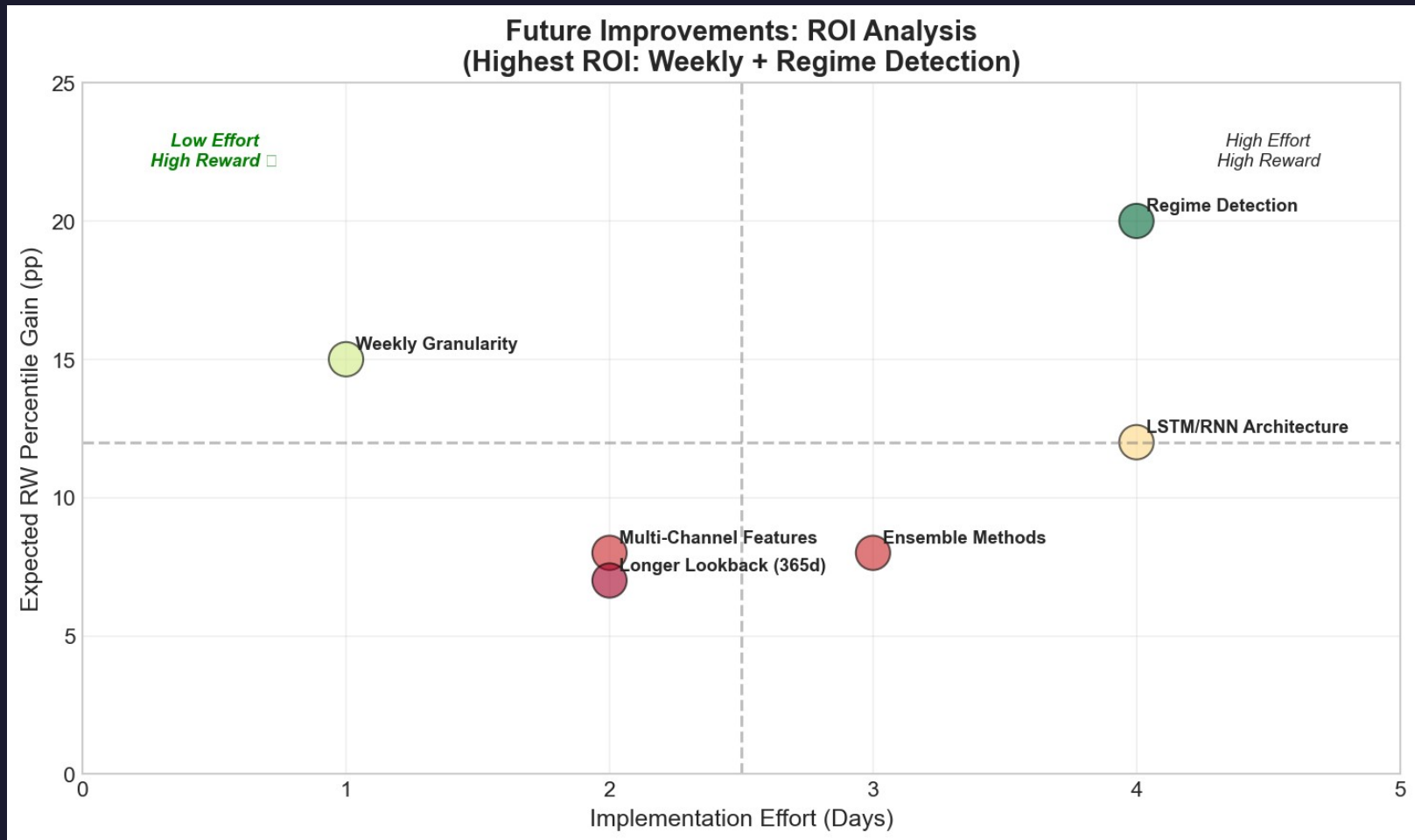
- Shift from 1-day direction to 5–10 day horizon signals
- Smooth/hold constant for weekly rebalancing

3. Robustness & Validation (Keep it Credible)

- Walk-forward evaluation with purged splits
- Report performance by subperiod + regime

Target: 41.94% → 57-73% RW percentile (to be validated)

Future Improvements & Next Steps



Hypothesis / goal range: 41.94% → 57-73% RW percentile (to be validated)

Risks & Mitigations

Regime Shifts: BTC market structure changed (institutional adoption)

Mitigation: Regime-aware model + rolling validation

Metric Brittleness: SPD sensitive to outliers

Mitigation: Multiple metrics, sanity gates implemented

Overfitting: Risk with regime detection on limited data

Mitigation: Strict train/test split, cross-validation

Data Quality: Exchange outages, feed gaps

Mitigation: Polygon as primary, validation checks

Data Leakage: Carefully validated, prevented

What We Will Deliver (Graded)

Reproducible Run:

- Single command execution
- Environment specification
- Deterministic output (seed=42)

Leakage/Causality Tests:

- Last-row test (PASS)
- Temporal embargo verified
- Zero lookahead confirmed

Models & Documentation:

- Midterm: Working baseline + all tests passing
- One improved model (weekly/regime)
- Final report + model card
- Ablation results table

Code Artifacts:

- Repository with README
- LESSONS_LEARNED.md
- Repro notebook (optional)

Midterm Result: We falsified a deep-learning approach under strict causality, pivoting to a simpler, more robust baseline

The Bottom Line

"We built a technically sophisticated system and discovered through rigorous testing that it performs statistically indistinguishable from random. This simplified version represents our evidence-based conclusion: in financial time series, signal quality matters more than model complexity."

Scientific Integrity > Raw Performance

Questions & Feedback Request

We'd appreciate feedback on:

1. Metric Interpretation: Is our SPD percentile understanding correct?
2. Split Strategy: Train (2014-2015) / Test (2016-2025) appropriate?
3. Pivot Scope: Is weekly + regime appropriately scoped?
4. Baseline Choice: Keep simplified or invest in XGBoost/ensemble?

Thank You

Questions?

Documentation: `LESSONS_LEARNED.md`

Code: `Visual_Trading_System` repository

Notebook: `btc_accumulation_model_simplified.ipynb`

"The best model is the one that works, not the one that sounds impressive."